FOUR EMBEDDEDNESS OBSTRUCTIONS IN GENUS-ONE FOUR-ENDED MINIMAL-SURFACE FAMILIES

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with the question of whether a complete properly embedded minimal surface in $\mathbb{R}^3$ of genus one can have four ends, we establish four exact family-specific embeddedness obstructions: (1) non-embeddedness of Costa's stated genus-one four-end Weierstrass family; (2) a genus/end lower bound in Traizet's balanced layered small-neck regeneration regime; (3) reduction of the normalized nonzero-spin reflection-symmetric $N_1 = N_2 = 0$, $N_3 = 2$ one-axis locus to the self-intersecting CCCC family; and (4) an Arf/spin parity obstruction for the Kato–Muroya zero-spin genus-one four-noid class.

1. Summary of the Four Obstructions

Theorem 1.1. *None of the following four construction families yields a complete properly embedded minimal surface in $\mathbb{R}^3$ of genus $g = 1$ with $k = 4$ ends:*

(1) Costa's elliptic four-end family: *closing the residues and periods at the four ends forces two catenoidal ends to share the same logarithmic growth rate or collide along the symmetry axis, destroying embeddedness.*

(2) Traizet's balanced small-neck configurations: *for $k = 4$ catenoidal layers with balanced neck charges, electrostatic equilibrium requires at least two independent vertical period balances, forcing $g \geq 2$ whenever embeddedness holds.*

(3) Normalized nonzero-spin $N_1 = N_2 = 0$, $N_3 = 2$ one-axis locus: *the negative half-period constraints force $\theta = \pi/2$ and reduce the Weierstrass data to the* CCCC *normal form, whose end ordering forces a self-intersection.*

(4) Kato–Muroya zero-spin four-noid class: *the Jorge–Meeks total curvature relation and Kusner–Mazzeo–Pollack spin/Arf invariant parity mismatch exclude an embedded representative.*

Date: August 29, 2026.

2020 *Mathematics Subject Classification.* Primary 53A10; Secondary 49Q05.